

RESIDUAL GAP CLASSES IN THE EVEN-DEGREE FERMAT-FOURFOLD CENSUS THROUGH DEGREE 250: EXCHANGE WALLS AND TWO CLOSURES

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ABSTRACT. For the Fermat fourfold $X_m^4: x_0^m + \cdots + x_5^m = 0$, the companion note [6] proves that the sector beyond the classical algebraicity machinery is empty in odd degree ≤ 199 . This paper is the systematic treatment of the *even*-degree sector through degree 250, which behaves differently: gap classes are frequent, and the odd law “survivor $\Rightarrow 3 \mid m$ ” fails. We first identify the working lattice with Aoki’s group S_m generator-by-generator — including his $p = 2$ standard elements, which we show are identically grade-2 Hodge quadruples, hence Lefschetz-algebraic — and prove a negation-closure lemma converting any integer membership certificate into a literal Aoki claim chain. The even-parity census — run under the note’s completeness lemma, which applies to both parities; receipts and independent verifiers in the ancillary — finds thirty-eight primitive beyond-machinery classes through degree 250; iterated partition certificates close all but ten. An exchange theorem generates claim-equivalences, and an exhaustive scan organizes the ten into five walls. Two of the five then close: the $m = 168$ class by Aoki’s published degree-form theorem (citing which we record two errata of the printed text, §5), and the isolated degree-210 class by an explicit 40-generator ± 1 certificate in S_{210} , displayed in full. The residual even-degree boundary at $m \leq 250$ — the classes unresolved by the implemented closure family and the cited published theorems — is eight certified gap classes ($\nu \notin S_m, 2\nu \in S_m$) in three exchange walls W_{70}, W_{110}, W_{114} ; every lattice verdict carries an independent certificate (modular kernel witnesses for non-membership, exact re-summation for membership).

1. INTRODUCTION

Let $X_m^4 \subset \mathbb{P}^5$ be the Fermat fourfold, $G = \mu_m^6/\mu_m$, and for a character $a = (a_0, \dots, a_5) \in (\mathbb{Z}/m)^6$ with $\sum a_i \equiv 0, a_i \neq 0$, let $V(a) \subset H_{\text{prim}}^4$ be the one-dimensional eigenspace; a is a Hodge $(2, 2)$ character iff $\sum_i \langle ta_i \rangle = 3m$ for all $t \in (\mathbb{Z}/m)^\times$ (Shioda [8]). The classical algebraicity machinery — Shioda’s inductive structure with da Silva’s quasi-decomposability formalism [8, 11, 5] and Aoki’s standard cycles [1] — and the notation of this paper are those of the companion note [6], whose conventions we use throughout: characters are *ordered* tuples; $\text{claim}(a)$ ($V(a)$ spanned by algebraic cycle classes) is invariant under coordinate permutations (automorphisms of X_m^4) and is a Galois-orbit-level property; the census classifies sorted representatives modulo both actions; *grade* g means $\sum_i \langle ta_i \rangle = gm$ for every unit t ; $*$ is juxtaposition of entry tuples, $\uplus$ multiset union; χ_a is the Jacobi-sum Hecke character of a and $u(a, p)$ its Tate-normalized value at a split prime p [12, 14], while $\nu(a) \in \mathbb{Z}^{m-1}$ is Aoki’s multiplicity vector, additive under juxtaposition (he writes u for ν); and the inflation/descent lemmas of [6] make claim level-transparent along $x \mapsto x^e$. The note proves (Theorems A, A', A'', B there) that every Hodge $(2, 2)$ class of every X_m^4 of *odd* degree $m \leq 199$ is algebraic.

This paper maps the even-degree sector through degree 250 and proves its structural results. The even sector is a genuinely different landscape: the first beyond-machinery classes appear at $m = 50$ (so “survivor $\Rightarrow 3 \mid m$ ”, true throughout the odd census, fails), most beyond-machinery

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classes close by *deeper* partition certificates rather than new mechanisms, and what finally remains is organized not by degree but by *claim-equivalence*. Our results:

- *The lattice is Aoki's* (Proposition 2.1): the working lattice — pair vectors and standard vectors, the $p = 2$ family included — coincides generator-by-generator with the group S_m of [3, §2], so all membership and gap verdicts are statements about Aoki's own object. The $p = 2$ standard elements are identically grade-2 Hodge quadruples, hence Lefschetz-algebraic (Lemma 2.2(i)), and the standard family is negation-closed modulo pairs (Lemma 2.2(ii)).
- *The census* (Proposition 3.1): thirty-eight primitive beyond-machinery classes through degree 250; iterated depth-3/4 partition certificates close all but ten.
- *Exchange walls* (Theorem 4.1, Proposition 4.2): a two-entry (or three-entry) exchange with a Lefschetz-algebraic exchanger is a claim-equivalence; the exhaustive scan organizes the ten residual classes into five walls.
- *Two closures* (Propositions 5.1 and 5.2): the $m = 168$ class falls under Aoki's published Theorem 0.1(i) [3] — we record two errata of the printed text, with a machine-verified possible replacement for the $m = 28$ gap-generator misprint — and the isolated degree-210 class closes by an explicit 40-generator ± 1 lattice certificate, displayed in full and converted to a literal Aoki chain.
- *The boundary* (Corollary 6.1): eight certified gap classes in three walls; closing any member of a wall closes the wall.

Status ledger: the Propositions labeled *computer-assisted* are exhaustive machine computations with pinned receipts, engines, and independent verifiers in the ancillary files; their scope (which levels, which move family) is stated in each. Everything else is proved. The obstruction theory of the closed odd classes (exact Frobenius certificates, the negative answer to da Silva's Question 1 at $m = 33$) is in [6] and not repeated here.

2. THE LATTICE IS AOKI'S S_m

S_m denotes Aoki's lattice: the subgroup of $\mathbb{Z}^{m-1}$ generated by the $\nu(\delta)$ over the vanishing pairs $\delta = \{k, m - k\}$ (the self-pair included at even m) and the $\nu(\sigma_{p,x})$ over the standard characters, for all primes $p \mid m$; membership is decided exactly by integer linear algebra (Hermite normal form), and in lattice statements we freely identify a with $\nu(a)$ — statements about the formal calculus, not about the character group. The first result pins the object.

Proposition 2.1 (the implemented lattice is Aoki's S_m). *Aoki [3, §2] defines A_m as the zero-sum sublattice of the free abelian group on $\mathbb{Z}/m \setminus \{0\}$; u — our ν — as the multiplicity-vector map, additive under juxtaposition, $\nu(\alpha * \beta) = \nu(\alpha) + \nu(\beta)$; the standard elements, verbatim, as $\sigma_{p,a} = (a, a + \frac{m}{p}, \dots, a + \frac{(p-1)m}{p}, m - pa)$ for $p \geq 3$ and $\sigma_{2,a} = (a, a + \frac{m}{2}, m - 2a, \frac{m}{2})$ for $p = 2$, over primes $p \mid m$ ($m > p$) and $0 < a < \frac{m}{p}$ with nonzero entries; $\mathfrak{S}_m$ as the set of α admitting $\alpha * \delta \sim \sigma_1 * \dots * \sigma_k * \delta'$ with δ, δ' unions of vanishing pairs, with $\mathfrak{D}_m \subset \mathfrak{S}_m$; and $S_m \subset A_m$ as the subgroup generated by $\nu(\mathfrak{S}_m)$. Consequently S_m equals the $\mathbb{Z}$ -span of the pair vectors $\nu(\{k, m - k\})$ and the standard vectors $\nu(\sigma_{p,a})$, $p = 2$ included — exactly the generator list of the implemented lattice — so the machine verdicts $\nu \in S_m / \nu \notin S_m$ are statements about Aoki's group itself.*

Proof. $\supseteq$: every pair class lies in $\mathfrak{D}_m \subset \mathfrak{S}_m$, and every standard element σ satisfies $\sigma * \delta \sim \sigma * \delta$, so $\sigma \in \mathfrak{S}_m$: each listed generator is in $\nu(\mathfrak{S}_m)$. $\subseteq$: if $\alpha * \delta \sim \sigma_1 * \dots * \sigma_k * \delta'$, then $\nu(\alpha) = \sum_i \nu(\sigma_i) + \nu(\delta') - \nu(\delta)$, an integer combination of the listed generators. The two spans coincide. $\square$

Lemma 2.2 ($p = 2$ standards; negation closure). (i) Every $\sigma_{2,x}$ is a grade-2 Hodge quadruple identically in x , so $V(\sigma_{2,x})$ is algebraic by Lefschetz (1,1) on the Fermat surface X_m^2 : the $p = 2$ generators carry the same unconditional algebraicity as the odd- p standards (Theorem 2-1 of [1], with the inflation lemma of [6] when $\gcd(x, m/p) > 1$). (ii) For every prime $p \mid m$ and admissible x , $\sigma_{p,x} \uplus \sigma_{p,m/p-x}$ is a union of vanishing pairs — $p+1$ of them for odd p , four for $p = 2$; in the lattice, $-\nu(\sigma_{p,x}) \equiv \nu(\sigma_{p,m/p-x}) \pmod{D_m}$, so an integer membership certificate converts to one with nonnegative standard coefficients at the cost of pair classes.

Proof. (i) For odd t write $r = \langle tx \rangle$; $t \cdot \frac{m}{2} \equiv \frac{m}{2}$, and the residues of $t \cdot \sigma_{2,x}$ are $r, r + \frac{m}{2}, m - 2r, \frac{m}{2}$ if $r < \frac{m}{2}$ and $r, r - \frac{m}{2}, 2m - 2r, \frac{m}{2}$ if $r > \frac{m}{2}$ ($r = \frac{m}{2}$ is excluded: $x + \frac{m}{2} \not\equiv 0$) — both sum to $2m$. Grade-2 Hodge quadruples are the (1,1) characters of X_m^2 ; the rational part of their Galois block is divisorial by Lefschetz (1,1), whence the claim. (ii) For odd p , $d = m/p$: the translates pair off as $(x + jd) + ((d - x) + (p - 1 - j)d) = pd = m$ for $j = 0, \dots, p - 1$, and $(m - px) + (px) = m$. For $p = 2$: $\{x, m - x\}$, $\{x + \frac{m}{2}, \frac{m}{2} - x\}$, $\{m - 2x, 2x\}$, and the self-pair $\{\frac{m}{2}, \frac{m}{2}\}$. Each pair lies in the span of the pair vectors. $\square$

Remark 2.3 (even- m standard admissibility). For even m and odd $p \mid m$ Aoki’s 1987 admissibility hypothesis $d/\gcd(x, d) > 2$ ($d = m/p$) is strictly stronger than “entries nonzero”: it excludes $x \equiv d/2 \pmod{d}$. Every excluded multiset contains a vanishing pair — $(x + k_1d) + (x + k_2d) \equiv 0 \pmod{m}$ reduces to $1 + 2j + k_1 + k_2 \equiv 0 \pmod{p}$ for $x = d/2 + jd$, solvable with $0 \leq k_1 < k_2 \leq p - 1$ — so it is decomposable, hence algebraic for the trivial reason, and the census classifier (which tests decomposability first) is unaffected. The lattice of Proposition 2.1 follows the range of [3, §2] (nonzero entries only), matching his definition of S_m .

Remark 2.4 (the gap group). Aoki’s Theorem 2.1 [3] (after [2, Thm. D]; the gap-group structure goes back to Yamamoto [13]) proves $2\alpha \in S_m$ for every Hodge class $\alpha \in B_m$: the gap group B_m/S_m is an elementary abelian 2-group, generated by level-raised gap generators ξ_d . Every “ $2u \in S_m$ ” certificate below therefore instantiates a general theorem; the content of a gap verdict is the non-membership half $u \notin S_m$.

3. THE EVEN CENSUS

For each even m : enumerate grade-3 Hodge multisets (meet-in-the-middle on half-unit sum profiles); reduce to Galois orbits of sorted representatives; classify decomposable $\rightarrow$ quasi-decomposable $\rightarrow$ standard $\rightarrow$ *-split (the transport theorem of [6], stated there for all m) $\rightarrow$ survivors. The census-completeness lemma of [6] is stated and proved there for both parities (the even self-pair $m/2$ included in $M_m(1)$); the even-parity engine is calibrated against direct brute force — exact match of representative counts for $m \leq 14$ — while the independent blind reimplementations of [6] cover the odd census. Throughout, *primitive* means content $\gcd(a_0, \dots, a_5, m) = 1$, i.e. not an inflation from a lower level; eigenvalue-field labels are *numerical observations* (an mpmath heuristic over finitely many split primes), used as descriptors only — no formal statement rests on them. Two engine-tier facts are recorded for transparency. The receipt engine (the v3 pipeline over the even-safe base library) is multiplicity-aware throughout: its decomposability requires both members of a vanishing pair, the self-paired $m/2$ requiring multiplicity two, and its quasi loop includes the self-pair $\{m/2, m/2\}$ (e.g. (1, 3, 36, 38, 64, 68) at $m = 70$ is genuinely quasi-decomposable through the self-pair, and the engine closes it). The odd-heritage display engine and the first version of the independent reimplementations carried set-based tests unsound at even m — a single $m/2$ reads as a vanishing pair; regression class (1, 29, 35, 43, 45, 57) at $m = 70$, neither decomposable nor quasi-decomposable. The independent reimplementations is fixed (multiplicity-aware), both regression classes are asserted afresh

by the runner, and its fresh key-level runs reproduce the engine receipt's representative and base-closure counts exactly at 50, 70, 110, 114, 168 (pinned receipt). Independently of any engine, an exhaustive self-pair-inclusive quasi test — every pair, every split, all units — certifies all ten residual-table classes non-quasi (pinned receipt): no wall is affected. For self-containedness we recall the imported transport statement: *a Hodge (2, 2) character of X_m^4 , m arbitrary, whose 6-multiset splits into two zero-sum triples has algebraic rational Hodge classes in its Galois block* ([6, Thm. A]; its transport uses only the Galois-stable block, Lefschetz (1, 1), and the $\mathbb{Q}$ -linear joining-line correspondence, so no rational point on the curve and no parity hypothesis enters). Counting convention: *beyond-machinery* counts the primitive orbits surviving all four base predicates (decomposable / quasi-decomposable / standard / *-split), before the iterated and deep tiers; the paper's $38 = 21 + 14 + 3$ is this base-tier primitive count over $m \leq 108$, $108 < m \leq 200$, $200 < m \leq 250$. Surviving orbits are then attacked by the *closure operator*: T closes at depth k if $T \uplus (k \text{ vanishing pairs})$, $k \leq 4$, partitions into blocks each of which is a vanishing pair, a grade-2 Hodge quadruple (Lefschetz (1, 1)), an admissible standard block (Theorem 2-1 of [1]; Lemma 2.2(i) at $p = 2$), or a previously certified sextuple (iteration: two-generation certificate trees); $\text{claim}(T)$ then follows by Theorems 1-4(i),(ii) of [1]. Every closure is stored as an explicit route in the deep-closure ledger and re-verified by the standalone checkers.

Proposition 3.1 (even census). (Computer-assisted; engines, closure ledger, and receipts in the ancillary.) (i) *For even $m \leq 108$ there are twenty-one primitive beyond-machinery orbits, beginning at $m = 50$, with indicated eigenvalue fields up to degree 32; the odd law “survivor $\Rightarrow 3 \mid m$ ” fails in the even sector.* (ii) *Eighteen of the twenty-one close by depth-3/4 partition certificates (two closed sextuples, or quadruple–standard–quadruple); iterating the closure operator (blocks certified by their own certificates — two-generation trees) closes $m = 54$ and $m = 90$ as well, so exactly one even class of degree ≤ 108 resists: $m = 70$, (1, 20, 24, 42, 61, 62), indicated eigenvalue field $\mathbb{Q}(\zeta_{35})$.* (iii) *The census through degree 200 yields fourteen further primitive classes, of which the same machinery closes eight (every open row in 178–200 is an induced copy of an already-closed class); the residual at $m \leq 200$ is seven classes (70; two at 110; three at 114; one at 168) and their induced copies.* (iv) *Through degree 250 exactly three more primitive classes appear: (2, 9, 129, 142, 168, 180) and (1, 79, 109, 121, 151, 169) at 210, and (1, 62, 111, 142, 162, 182) at 220.*

Among the eighteen closures of (ii) is the $m = 66$, $\mathbb{Q}(\sqrt{-3})$ class — closed through the self-paired element (33, 33), the same resource that closes da Silva's $m = 33$ witness at level 66 in [6, Thm. A''] — so at that census depth every known Fermat-fourfold Hodge class with imaginary-quadratic *indicated* eigenvalue field was algebraic; the deeper census produces a third such class at $m = 168$, closed in §5. That survivor, (1, 25, 79, 121, 127, 151), has indicated eigenvalue field $\mathbb{Q}(\sqrt{-3})$, and its Frobenius character has a kernel prime ($p = 673$), so the field-of-definition obstruction of [6, Thm. C] vanishes there.

4. EXCHANGE AND THE WALL STRUCTURE

Theorem 4.1 (Exchange). *Let $T = S \uplus A$ and $T' = S \uplus B$ be Hodge sextuples mod m (entries nonzero) with $|A| = |B| = 2$, and suppose $q := A \uplus (-B)$ has nonzero entries and is a Hodge (1, 1) quadruple. Then $\text{claim}(T') \Rightarrow \text{claim}(T)$; since $-q$ serves the reverse direction, T and T' are claim-equivalent. The same holds for $|A| = 3$ with q a sextuple whose claim is known.*

Proof. $V(q)$ is algebraic by Lefschetz (1, 1) on the Fermat surface X_m^2 (no membership in $\mathfrak{D}_m$ is required). By Theorem 1-4(i) of [1] (juxtaposition join, r, s even), $\text{claim}(T')$ and $\text{claim}(q)$ give $\text{claim}(T' * q)$. As multisets, $T' \uplus q = T \uplus (y_1, -y_1) \uplus (y_2, -y_2) = T * \delta$ with $\delta = (y_1, -y_1, y_2, -y_2)$ decomposable, so $\delta \in \mathfrak{D}_m^2$ and Theorem 1-4(ii) of [1] cancels it. For $|A| = 3$ the same two steps

apply with q a sextuple of known claim and $\delta = B \uplus (-B)$ three vanishing pairs, $\delta \in \mathfrak{D}_m^4$ ($s = 4$ even); and $\text{claim}(-q) = \text{claim}(q)$ since $-q = (-1) \cdot q$ is a Galois conjugate. (Landing on a Galois conjugate $t \cdot T'$ suffices, claim being orbit-level; the $(m/2, m/2)$ self-pair is a legitimate multiset entry.) $\square$

Proposition 4.2 (wall structure through degree 250). (Computer-assisted: exhaustive exchange scan, both $|A|$ shapes, own levels and level-lifts.) *The two $m = 110$ classes are claim-equivalent — the $\mathbb{Q}(\zeta_5)$ class stands or falls with the $\mathbb{Q}(\zeta_{110})$ one — and the three $m = 114$ classes form a complete triangle at Galois-orbit level, while $m = 70$ and $m = 168$ are exchange-isolated. The three primitive classes of Proposition 3.1(iv) are already placed by the exchange graph: $(2, 9, 129, 142, 168, 180)$ at 210 is equivalent to the 3-lift of the $m = 70$ class; the 220 class is equivalent to the 2-lifts of the $m = 110$ pair; $(1, 79, 109, 121, 151, 169)$ at 210 is new and isolated. The residual boundary after the census machinery — before the closures of Propositions 5.1 and 5.2 — is thus ten primitive classes organized in five claim-equivalence walls*

$$\begin{aligned} W_{70} &= \{70 \simeq 210\#2\}, & W_{110} &= \{110\#1 \simeq 110\#2 \simeq 220\}, \\ W_{114} &= \{\#1 \simeq \#2 \simeq \#3\}, & W_{168}, & & W_{210} &= \{210\#1\}, \end{aligned}$$

and closing any member of a wall closes the wall (Theorem 4.1). Precisely: call $T \leftrightarrow T'$ an exchange move when $T = S \uplus A$, $T' = S \uplus B$ with $|A| = |B| \in \{2, 3\}$ and the exchanger $q = A \uplus (-B)$ has nonzero entries and is a grade-2 Hodge quadruple ($|A| = 2$; unconditional by Lefschetz) or a grade-3 sextuple of known claim ($|A| = 3$). The scan enumerated every move out of every open class — all entry positions, all B — at its own level and its lifts within the census range (pinned receipts: the own levels 70, 110, 114, 168, 210 and the lifted levels 140, 220, 228); the open set supplied to the scan at each level is the complete final list of unresolved orbits there, so an $|A| = 3$ exchanger outside it carries a known algebraicity claim; each lands on the class's own wall, never on a closed class. Within this move family and level range, the transitive exchange closure of each open class is therefore exactly its wall.

5. TWO CLOSURES

Proposition 5.1 (the $m = 168$ class closes). *The class $(1, 25, 79, 121, 127, 151)$ on X_{168}^4 is algebraic; with it every class of the two census ranges (odd $m \leq 199$, even $m \leq 250$) whose indicated eigenvalue field is imaginary quadratic ($m = 33, 66, 168$) is closed.*

Proof. The closure rests on the published theorem: Aoki [3] proves (Theorem 0.1(i)) the Hodge conjecture for all abelian varieties of Fermat type of degree $m = 2^a 3^b 5^c 7^d$ ($c = 0$ or $d = 0$), and the proof's §4 route opens with the bridge through Corollary 3.2 there, establishing the statement for the Fermat varieties X_m^n themselves, for all n , at these degrees; $168 = 2^3 \cdot 3 \cdot 7$ qualifies. We do not re-derive that proof; we record two errata of the printed text, both verified against the primary and neither affecting the theorem as stated. First, the paper's own later citation of this statement ([4, Thm. 1.2(ii)], attributed to a then-preprint on Catalan curves) is a citation slip — the published Catalan paper (Tokyo J. Math. **27** (2004)) concerns the Jacobians of generalized Catalan curves, not the degree form; the theorem is [3]'s own Thm. 0.1(i). Second, in the §4 list of gap generators the $m = 28$ entry $\xi_{28} = (1, 9, 25, 12, 20, 24)$ is a misprint (it is not a zero-sum character), while Aoki's own text requires $\xi_m \in B^2 \cup (B^4 \cap (\mathfrak{A}^1 * \mathfrak{A}^1))$ and algebraizes these $V(\xi_m)$ by the proof of Theorem 4.3 of Shioda [9] (cited there); a machine-verified possible replacement of exactly the required join form is $\xi_{28} = (1, 9, 18) * (10, 21, 25) \in B_{28}^4 \cap (\mathfrak{A}_{28}^1 * \mathfrak{A}_{28}^1)$, $\nu(\xi_{28}) \notin S_{28}$, $2\nu(\xi_{28}) \in S_{28}$ (whether it is the intended generator modulo S_{28} is not established here; the closure of 168 does not depend on it). At the class level our lattice computation gives $\nu(a) \notin S_{168}$ with $\nu(a) \equiv \nu(6 \cdot \xi_{28}) \bmod S_{168}$ (entrywise product — the level-raising map of [3, §2]): the class is a genuine gap class, and no Fermat-internal route of this paper or of [6] reaches

it (its multiset admits no zero-sum-triple split, so the $*$ -split theorem — stated in [6] for all m — does not fire, machine-checked; $5 \nmid 168$ leaves no standard block for the two-pair genre; the iterated and lift searches also failed) — Theorem 0.1(i) is what closes it. $\square$

Proposition 5.2 (the isolated degree-210 class closes). $V(1, 79, 109, 121, 151, 169)$ on X_{210}^4 is algebraic.

Proof. The class closes by an explicit certificate: $\nu(1, 79, 109, 121, 151, 169) \in S_{210}$, realized by 40 generators with coefficients ± 1 (re-summed independently in the ancillary; stored as a data artifact). In full, with (k) denoting the pair $\{k, 210 - k\}$:

$$\begin{aligned} \nu(a) = & -(7)-(8)+(12)-(21)-(27)-(28)-(29)+(36)+(41)+(42)-(61) \\ & +(66)+(74)-(76)-(77)-(78)+(82)+(84)-(91)+(96)-(105) \\ & +\sigma_{2,4}-\sigma_{2,9}+\sigma_{2,14}+\sigma_{2,21}+\sigma_{2,27}+\sigma_{2,28}+\sigma_{2,29}-\sigma_{2,31}-\sigma_{2,41}-\sigma_{2,42} \\ & -\sigma_{3,4}+\sigma_{3,6}+\sigma_{3,7}+\sigma_{3,8}+\sigma_{3,9}-\sigma_{3,12}-\sigma_{3,14}+\sigma_{7,1}-\sigma_{7,6}. \end{aligned}$$

Every generator's eigenspace is algebraic — pairs: linear cycles, Theorem 1-1 of [1]; $\sigma_{3,x}, \sigma_{7,x}$: Theorem 2-1 of [1], with the inflation lemma of [6] when $\gcd(x, m/p) > 1$; $\sigma_{2,x}$: Lemma 2.2(i) — so converting the negative terms by Lemma 2.2(ii) produces the literal membership $a * \delta \sim \sigma_1 * \dots * \sigma_{19} * \delta'$ (δ, δ' unions of vanishing pairs), and $V(a)$ is algebraic by Theorems 1-4(i),(ii) of [1] — the calculus of Lemma 4.1 of [3] (which carries no hypothesis on m), with every block certified above. $\square$

Negative-coefficient certificates of this kind are invisible to monoid/partition search engines — the lattice membership test ($\nu(a) \in S_m$, Hermite normal form) decides standard-calculus reachability exactly and is the instrument we now run first. (The two directions, with the scope stated precisely: a certificate all of whose blocks are vanishing pairs, standard blocks, or classes already certified inside S_m forces $\nu(a) \in S_m$, since ν is additive under $*$; a certificate routed through a Lefschetz quadruple whose own ν lies outside S_m escapes the lattice — the $\mathbb{Q}(\zeta_{21})$ class at 105 closes exactly this way in [6, Thm. A'] while remaining a gap class — so gap-ness bounds the *standard* calculus, not partition search as such. Conversely an integer membership combination converts to a literal Aoki chain by negation closure (Lemma 2.2(ii)), as carried out in Proposition 5.2.)

6. THE BOUNDARY AT $m \leq 250$

Corollary 6.1 (the even-degree boundary at $m \leq 250$). *The residual boundary for all even degrees $m \leq 250$ — unresolved by the implemented closure family and the cited published theorems — is eight primitive classes in three walls (W_{70} : two members, W_{110} : three, W_{114} : three, including the displayed lift-equivalent members at 210 and 220), and all eight are certified gap classes: $\nu \notin S_m$, $2\nu \in S_m$ (the doubling holds for every Hodge class by [3, Thm. 2.1]; the certificates make it explicit). The residual question is uniform: algebraicity of gap-group classes at degrees 70, 110, 114 — outside the degree forms of Aoki's theorem (70 and 210 contain both 5 and 7; 110 contains 11; 114 contains 19).*

m	class	$\nu \in S_m?$	status
70	(1, 20, 24, 42, 61, 62)	no	unresolved (W_{70})
110	(1, 24, 62, 71, 81, 91)	no	unresolved (W_{110})
110	(1, 31, 55, 71, 81, 91)	no	unresolved (W_{110})
114	(1, 13, 43, 72, 103, 110)	no	unresolved (W_{114})
114	(1, 13, 43, 80, 102, 103)	no	unresolved (W_{114})
114	(1, 7, 78, 79, 86, 91)	no	unresolved (W_{114})
168	(1, 25, 79, 121, 127, 151)	no	closed (Prop. 5.1)
210	(1, 79, 109, 121, 151, 169)	yes	closed (Prop. 5.2)
210	(2, 9, 129, 142, 168, 180)	no	unresolved (W_{70})
220	(1, 62, 111, 142, 162, 182)	no	unresolved (W_{110})

The ten primitive classes of degree ≤ 250 beyond the base machinery. Every lattice verdict carries an independent certificate in the ancillary: a modular kernel witness ($q = 2$) for each “no”, the re-summed ± 1 combination for the “yes”.

A systematic treatment of the even sector beyond degree 250 is deferred.

7. METHODS, CODE AVAILABILITY, AND PROVENANCE

Exact objects throughout: multiset identities, Hodge grades, and lattice membership are decided in exact integer arithmetic; no floating point enters any certificate. Every lattice verdict has an independent exact certificate (modular kernel witnesses found by the standalone script’s own elimination for each non-membership; exact re-summation for each membership and for the doubled vectors); the full census and exchange sweeps are reproducible from the supplied engines and are accompanied by checksummed campaign receipts; independent brute-force calibration is performed at small levels ($m \leq 14$), and the algorithmically independent reimplementations re-derives the representative counts fresh at the key levels 50, 70, 110, 114, 168 (pinned receipt). The exact-arithmetic certificate framework is the one developed for the Fermat surfaces of [7], and the odd-degree half of the census is certified in [6] by three independent implementations, including a direct brute-force enumeration at every level $m \leq 143$. The author takes full responsibility for all proofs, code, and bibliographic claims. The ancillary package is shared with the companion note: engines (census, closure oracle, exchange scan, lift sweeps, lattice membership), the deep-closure ledger, the stored degree-210 certificate, the complete even census receipt over all even $6 \leq m \leq 250$, and the pinned exchange/negative-sweep logs, with a checksum manifest and a fresh-run runner (`run_all.sh`; “PASSED” always means executed now, never cached). This manuscript was prepared with AI assistance; all computational claims are backed by the ancillary code and receipts.

APPENDIX A. THE CITED AOKI STATEMENTS

For self-containedness we reproduce the exact statements used, in the numbering of the primary texts.

From [1] (notation as there: $\mathfrak{B}_m^n$ the Hodge characters of X_m^n , $\mathfrak{D}_m^n$ the classes that are coordinate permutations of $(a_0, -a_0, \dots, a_r, -a_r)$, $r = n/2$; $*$ juxtaposition; CLAIM(α): $V(\alpha)$ is generated by classes of algebraic cycles on X_m^n):

- **Theorem 1-1** (Shioda). *The linear space L represents $\delta \in \mathfrak{D}_m^n$; more precisely $\omega_\delta(L) \cdot \overline{\omega_\delta(L)} = (-1)^r m^{n+1}$.*
- **Theorem 1-4** (Shioda [8], Ran [10]). *Let r and s be non-negative even integers such that $n = r + s + 2$, and let $\alpha \in \mathfrak{B}_m^r$, $\beta \in \mathfrak{B}_m^s$. Then: (i) if CLAIM(α) and CLAIM(β) are true, then CLAIM($\alpha * \beta$) is also true; (ii) if there exists $\delta \in \mathfrak{D}_m^s$ such that CLAIM($\alpha * \delta$) is true, then CLAIM(α) is also true.*

- **Theorem 2-1.** *For an odd prime $p \mid m$, $d = m/p$, $\gcd(a, d) = 1$, the variety Y defined by (2.1) there is a subvariety of X_m^{p-1} of codimension $r = (p-1)/2$ representing $\alpha = \sigma_{p,a}$; more precisely $\omega_\alpha(Y) \cdot \overline{\omega_\alpha(Y)} = (-1)^r p^{p-2} m^p$.*

From [3] (verified verbatim against the open-access copy in the Rikkyo University repository, doi:10.14992/00009819; §5 records two errata of the printed text, both likewise verified against the primary):

- **§2 definitions** (verbatim). *If m is divisible by a prime number p and $m > p$, then we have explicitly defined elements of $\mathfrak{B}_m$: $\sigma_{p,a} = (a, a + \frac{m}{p}, \dots, a + \frac{(p-1)m}{p}, m - pa)$ if $p \geq 3$; $(a, a + \frac{m}{2}, m - 2a, \frac{m}{2})$ if $p = 2$ — where a is an integer such that $0 < a < \frac{m}{p}$. We call $\sigma_{p,a}$ a standard element. Further (§2 there): A_m is the zero-sum sublattice of the free abelian group on $\mathbb{Z}/m \setminus \{0\}$; u (our ν) the multiplicity-vector map, $\nu(\alpha * \beta) = \nu(\alpha) + \nu(\beta)$; $B_m, S_m, D_m \subset A_m$ the subgroups generated by $u(\mathfrak{B}_m), u(\mathfrak{S}_m), u(\mathfrak{D}_m)$, with $\mathfrak{D}_m \subset \mathfrak{S}_m \subset \mathfrak{B}_m$ and $\mathfrak{S}_m$ the set of α admitting $\alpha * \delta \sim \sigma_1 * \dots * \sigma_k * \delta'$ ($\delta, \delta' \in \mathfrak{D}_m$).*
- **Theorem 0.1.** *Suppose the prime factorization of m is one of the following forms: (i) $m = 2^a 3^b 5^c 7^d$, where a, b, c, d are non-negative integers such that either $c = 0$ or $d = 0$; (ii) $m = p^e$ or $2p^e$, p an odd prime. Then the Hodge conjecture is true for all abelian varieties of Fermat type of degree m . Its proof (§4 there, p. 185) opens: “In view of Corollary 3.2, it suffices to show that the Hodge conjecture for X_m^n is true for all n ” — the bridge to the Fermat varieties themselves used in Proposition 5.1; the verification then reduces, via Lemma 4.1 and Theorem 2.1 there, to the gap generators ξ_m at $m = 12, 15, 20, 21, 28$ (the $m = 28$ entry being the misprint recorded in §5).*
- **Theorem 2.1** (verbatim; proved as Theorem D of [2]). *For each divisor $d > 1$ of m such that $\mu'(d) = 1$, there exists an element ξ_d of $B_d - S_d$ with the property that the Γ_m -module B_m is generated by $(m/d)\xi_d$ for all such divisors d of m and S_m . Moreover, we have $2\alpha \in S_m$ for every $\alpha \in B_m$. (μ' is Aoki’s modified Möbius function and (m/d) his level-raising product map; their exact forms are not needed here.)*
- **Lemma 4.1** (verbatim). *Let $\alpha, \beta \in B_m$. Then: (i) if both $V(\alpha)$ and $V(\beta)$ are algebraic, then so is $V(\alpha * \beta)$; (ii) if $V(\alpha * \delta)$ is algebraic for some $\delta \in D_m$, then so is $V(\alpha)$; (iii) if $\alpha \in S_m$, then $V(\alpha)$ is algebraic. No hypothesis on m is imposed; Proposition 2.1 identifies the implemented lattice with S_m generator-by-generator.*

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